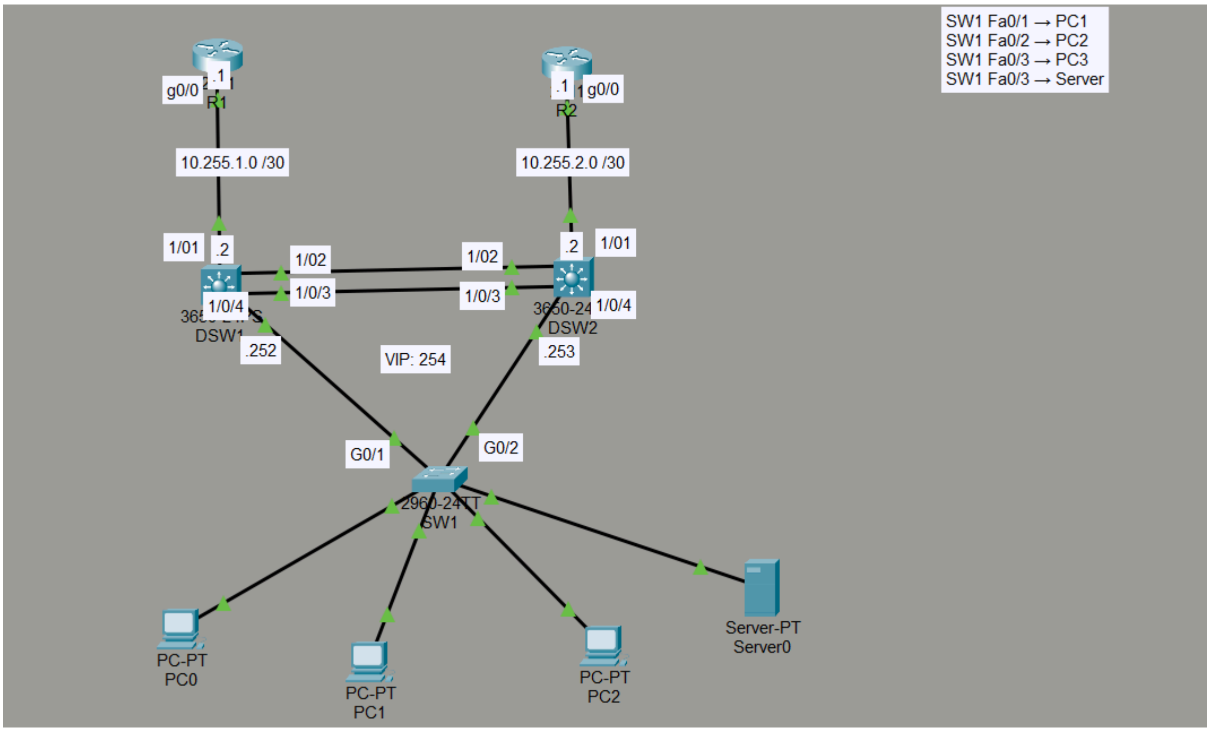
CCNA Progressive Lab — HSRP Gateway Redundancy

Scenario

The company has two Layer 3 distribution switches. Previously, each VLAN had a single default gateway, creating a **single point of failure**.

The task is to implement **HSRP** so that users have a redundant default gateway.



Physical Connections

Use these exact connections.

R1 → DSW1

R1 G0/0 ↔ DSW1 G1/0/1

R2 → DSW2

R2 G0/0 ↔ DSW2 G/10/1

DSW1 → DSW2

Use **two physical links**:

DSW1 G1/0/2 ↔ DSW2 G1/0/2
DSW1 G1/0/3 ↔ DSW2 G1/0/3

These will become an **LACP EtherChannel**.

DSW1 → SW1

DSW1 G1/0/4 ↔ SW1 G0/1

DSW2 → SW1

DSW2 G1/0/4 ↔ SW1 G0/2

End devices

SW1 Fa0/1 → PC1
SW1 Fa0/2 → PC2
SW1 Fa0/3 → PC3

Server:

SW1 Fa0/4 → Server0

VLANs

Create the following VLANs:

VLAN	Name	Network	HSRP Virtual Gateway
10	USERS	10.10.10.0/24	10.10.10.254
20	ADMIN	10.10.20.0/24	10.10.20.254
30	SERVERS	10.10.30.0/24	10.10.30.254
99	MANAGEMENT	10.10.99.0/24	10.10.99.254

In Switch 1:
Vlan 10
Name users
Vlan 20
Name admin
and like that until all Vlans get created.

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   users                            active    Fa0/1, Fa0/3
20   admin                            active    Fa0/2
30   servers                          active    Fa0/4
99   management                       active
```

HSRP Addressing

For every VLAN, DSW1 and DSW2 will have their **own physical SVI address**, while HSRP provides the **virtual gateway**.

VLAN 10:
DSW1: 10.10.10.252

- int vlan 10: Create the SVI
- ip add 10.10.10.252 255.255.255.252: Adding an IP address to the SVI

DSW2: 10.10.10.253

- int vlan 10: Create the SVI
- ip add 10.10.10.253 255.255.255.252: Adding an IP address to the SVI

HSRP: 10.10.10.254

- standby 10 ip 10.10.10.254: Creates the HSPR group and assign the virtual IP.

VLAN 20:
DSW1: 10.10.20.252

- int vlan 20: Create the SVI
- ip add 10.10.20.252 255.255.255.252: Adding an IP address to the SVI

DSW2: 10.10.10.253

- int vlan 20: Create the SVI
- ip add 10.10.20.252 255.255.255.252: Adding an IP address to the SVI

HSRP: 10.10.10.254

- standby 20 ip 10.10.20.254: Creates the HSPR group and assign the virtual IP

And like this until it's done for all Vlans. This needs to be done from the interface configuration mode.

HSRP Roles

Configure the following intentionally:

VLAN	Active Router	Standby Router
VLAN 10	DSW1	DSW2
VLAN 20	DSW2	DSW1
VLAN 30	DSW1	DSW2
VLAN 99	DSW2	DSW1

In DSW1:

- **int vlan 10:**
- standby 10 priority 50: Give priority to the group (lower the priority will be selected as active)
- standby 10 preempt: Overthrow lower priority Active routers.
- **int vlan 30:**
- standby 30 priority 150: Give priority to the group (lower the priority will be selected as active)
- standby 30 preempt: Overthrow lower priority Active routers.

In DSW2:

- **int vlan 20:**
- standby 20 priority 50: Give priority to the group (lower the priority will be selected as active)
- standby 20 preempt: Overthrow lower priority Active routers.

- **int vlan 99:**
- standby 99 priority 50: Give priority to the group (lower the priority will be selected as active)
- standby 99 preempt: Overthrow lower priority Active routers.

It should look like this

```
DSW1#sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   200 P Active  local           10.10.10.253    10.10.10.254
Vl20        20   50    Standby 10.10.20.253    local           10.10.20.254
Vl30        30   200 P Active  local           10.10.30.253    10.10.30.254
Vl99        99   50  P Standby 10.10.99.253    local           10.10.99.254
DSW1#
```

```
DSW2#sh standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        10   50    Standby 10.10.10.252    local           10.10.10.254
Vl20        20   200 P Active  local           10.10.20.252    10.10.20.254
Vl30        30   50    Standby 10.10.30.252    local           10.10.30.254
Vl99        99   200 P Active  local           10.10.99.252    10.10.99.254
DSW2#
```

PC Addressing

PC1 — Users
IP: 10.10.10.10
Mask: 255.255.255.0
Gateway: 10.10.10.254

PC2 — Admin
IP: 10.10.20.10
Mask: 255.255.255.0
Gateway: 10.10.20.254

PC3 — Users
IP: 10.10.10.11
Mask: 255.255.255.0
Gateway: 10.10.10.254

Server0
IP: 10.10.30.10
Mask: 255.255.255.0
Gateway: 10.10.30.254

DSW1 ↔ DSW2 EtherChannel

Create an **LACP EtherChannel** using:

DSW1 G1/0/2
DSW1 G1/0/3

DSW2 G1/0/2
DSW2 G1/0/3

Use:

Port-Channel 1

- int ra G1/0/2-3: Select the interfaces the will participate in the channel group
- channel-group 1 mode active: Create the channel group

The Port-Channel should operate as a **trunk** and carry:

VLAN 10
VLAN 20
VLAN 30
VLAN 99

- int po1: Enter the port-channel interface
- no shutdown: turn the interface up
- sw mo tr: Change the interface to trunk mode
- sw tr allow vlan 10,20,30,99: Add the Vlans to the trunk port.

Verify with:

show etherchannel summary

```
DSW1#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/2(P) Gig1/0/3(P)
DSW1#
```

and:

show interfaces trunk

```
DSW1#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Gig1/0/2(P) Gig1/0/3(P)
DSW1#sh int tr
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/4    auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Po1         10,20,30,99
Gig1/0/4    1-1005

Port        Vlans allowed and active in management domain
Po1         10,20,30,99
Gig1/0/4    1,10,20,30,99

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,99
Gig1/0/4    1,10,20,30,99

DSW1#
```

DSW1 / DSW2 → SW1

The links between the distribution switches and SW1 must also be **802.1Q trunks**.

Allow: VLAN 10 VLAN 20 VLAN 30 VLAN 99

The important concept is that both DSW1 and DSW2 need Layer 2 access to the VLANs because HSRP peers must communicate through the same VLAN.

SW1

- int ra g0/1-2: Select the interface range
- sw mo tr: Change the interface to trunk mode
- sw tr allow vlan 10,20,30,99: Add the Vlans to the trunk port.

DSW

- int g1/0/4: Select the interface range
- sw mo tr: Change the interface to trunk mode
- sw tr allow vlan 10,20,30,99: Add the Vlans to the trunk port.

DSW

- int g1/0/4: Select the interface range
- sw mo tr: Change the interface to trunk mode
- sw tr allow vlan 10,20,30,99: Add the Vlans to the trunk port.

OSPF

Now connect theHSRP distribution layer to the routed network.

Use:

R1 ↔ DSW1

R1 = 10.255.1.1 DSW1 = 10.255.1.2

On R1

- router ospf 1: Enter into the router configuration mode
- net 10.255.1.0 0.0.0.3 area 0: Advise this network.

On DSW:

- router ospf 1: Enter into the router configuration mode
- net 10.10.10.0 0.0.0.255 area 0: Advise this network.
- net 10.10.20.0 0.0.0.255 area 0: Advise this network.
- net 10.10.30.0 0.0.0.255 area 0: Advise this network.
- net 10.10.99.0 0.0.0.255 area 0: Advise this network.

R2 ↔ DSW2

R2 = 10.255.2.1 DSW2 = 10.255.2.2

On R2

- router ospf 1: Enter into the router configuration mode
- net 10.255..2.0 0.0.0.3 area 0: Advise this network.

On DSW:

- router ospf 1: Enter into the router configuration mode
- net 10.10.10.0 0.0.0.255 area 0: Advise this network.
- net 10.10.20.0 0.0.0.255 area 0: Advise this network.
- net 10.10.30.0 0.0.0.255 area 0: Advise this network.
- net 10.10.99.0 0.0.0.255 area 0: Advise this network.

```
DSW1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
22.22.22.22       1   FULL/DR         00:00:39    10.10.10.253    Vlan10
22.22.22.22       1   FULL/BDR        00:00:31    10.10.30.253    Vlan30
22.22.22.22       1   FULL/DR         00:00:38    10.10.20.253    Vlan20
22.22.22.22       1   FULL/DR         00:00:38    10.10.99.253    Vlan99
1.1.1.1           1   FULL/BDR        00:00:38    10.255.1.1
GigabitEthernet1/0/1
DSW1#
```

What I understood by the end?

The PC doesn't know or care which physical switch is currently Active. That's the core purpose of FHRP/HSRP: default-gateway redundancy.